TECHNICAL PROJECT REPORT

Visualec

AI-Powered Vision-Based Adaptive Grid Energy Management System

INT8 edge inference, zone-wise occupancy control, relay actuation, monitoring, and measured ESP32-S3 performance.

PROJECT OBJECTIVE

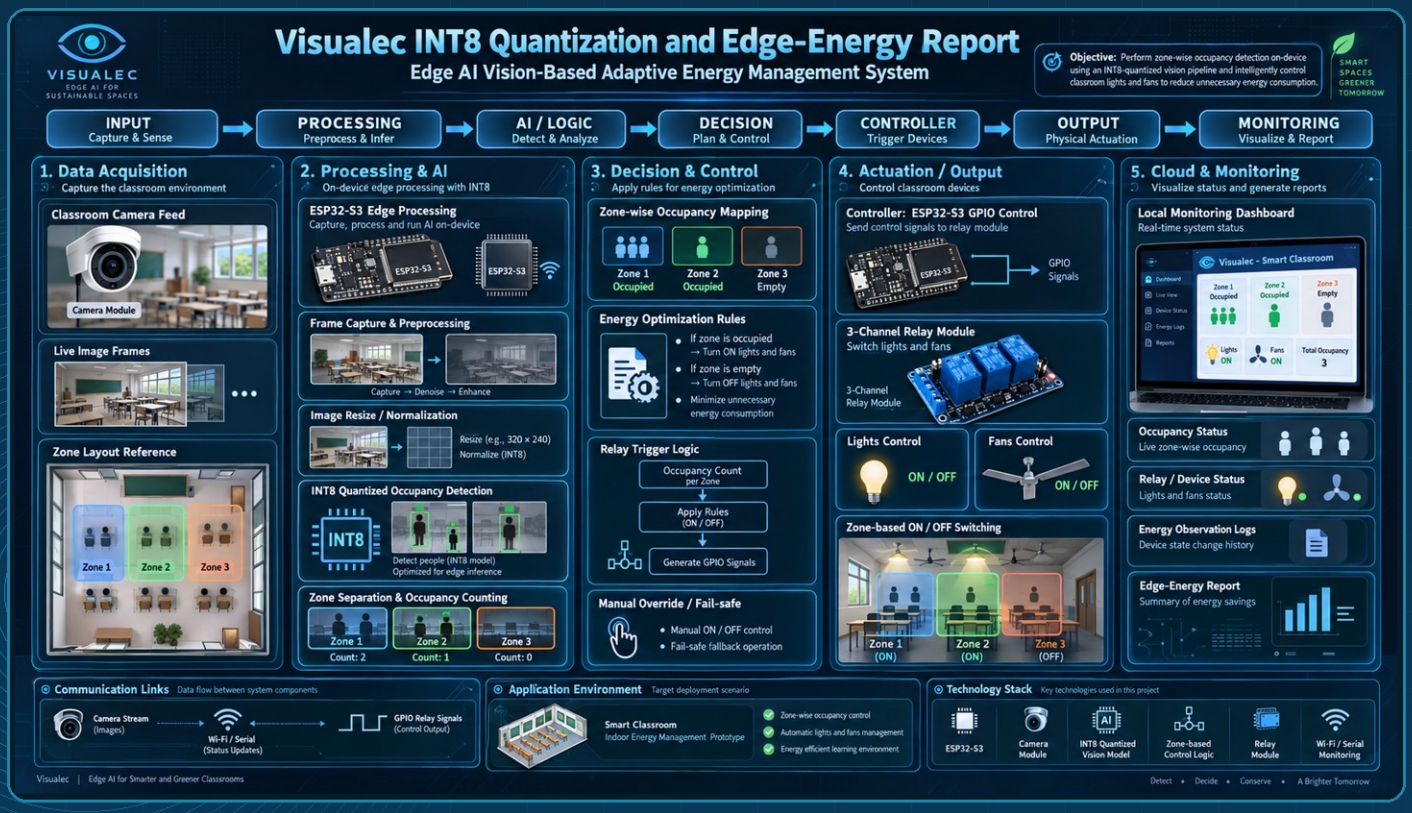
Perform zone-wise occupancy detection using an INT8 edge-vision pipeline and intelligently control classroom lights and fans to reduce unnecessary energy consumption.

8 FPS
Observed compute capacity

126 ms
Observed device inference

299 ms
Observed round-trip latency

INT8
w8a8 deployment path



Evidence snapshot: 06 Sep 2026

Version 1.0

Document Control & Abstract

Field	Details
Project	Visualec: An AI-Powered Vision-Based Adaptive Grid Energy Management System
Document	Complete Technical Project Report - INT8 Edge AI Prototype
Report version	1.0
Evidence date	06 September 2026
Primary deployment target	ESP32-S3 N16R8 edge device
Prototype environment	Zone-based smart classroom energy management
Evidence boundary	Measured device inference and runtime evidence; energy savings not yet power-meter verified

Abstract

Visualec is a vision-based adaptive energy-management prototype designed to control classroom loads according to zone-wise human occupancy. The system separates the monitored room into logical zones, performs person detection through an INT8 edge-inference pipeline, maps detections to zones, applies occupancy and hysteresis rules, and controls lights or fans through relay outputs. The current physical evidence demonstrates real INT8 pedestrian inference on an ESP32-S3 using Espressif ESP-DL. The observed sample achieved approximately 126 ms device inference latency and 8 FPS compute capacity, while end-to-end round-trip latency was approximately 299 ms. The report also documents memory utilization, the current bootstrap detector, the planned YOLOv8n-to-compact-student distillation path, relay safety behavior, communication design, dashboard monitoring, and the methodology required before claiming physical energy savings.

> **Evidence-first reporting**
> The report deliberately separates what has been measured on the physical ESP32-S3 from what remains planned. The custom Visualec student model and building-level energy savings are not presented as completed results.

Report Contents

- 1. Executive Summary
- 2. Problem Statement, Objectives & Scope
- 3. System Architecture
- 4. Functional Workflow
- 5. Hardware Design
- 6. Software & AI Architecture
- 7. INT8 Quantization & Model Pipeline
- 8. Firmware Scheduling, Safety & Control Logic
- 9. Communication & Monitoring
- 10. Performance Validation
- 11. Memory & Partition Analysis
- 12. Energy Management & Measurement Methodology
- 13. Implementation Status, Risks & Limitations
- 14. Scalability & Future Roadmap
- 15. Sustainability Alignment
- 16. Conclusion
- Appendix A - Verification Matrix
- Appendix B - Key Parameters & Glossary
- References

OVERVIEW

1. Executive Summary

The Visualec prototype targets a common source of avoidable energy consumption in classrooms: lighting and ventilation loads remaining active even when only part of a room is occupied. Instead of treating an entire classroom as one binary occupied/unoccupied space, Visualec divides the room into zones and independently controls the associated loads.

The current implementation uses an ESP32-S3 as the edge inference and control device. Frames are prepared upstream, transported in RGB565 format, processed with an INT8 pedestrian detector, and converted into zone-wise occupancy decisions. Control decisions are translated into active-low GPIO states that operate relay channels. A local dashboard presents occupancy, relay status, logs, and software-derived energy observations.

8 FPS	126 ms	299 ms	0
Compute capacity	Inference latency	Round trip	Protocol errors

Current acceptance status

Device inference meets the compute gate: 8 FPS exceeds the 7 FPS floor and 126 ms is below the 140 ms inference ceiling. The observed 299 ms round trip does not meet the 200 ms p95 target and remains the primary performance gap.

Key engineering findings

- INT8 reduces scalar storage to 25% of FP32 for the same tensor count, improving fit and reducing data movement; this does not imply a guaranteed 4x reduction in latency or power.
- The deployed physical model is the ESP-DL pedestrian-pico-bootstrap detector, not the planned custom visualec_pico.espdl model.
- The built model occupies 435,440 bytes (20.76% of its 2 MiB partition), leaving substantial model-space headroom.
- The application binary occupies 2,865,264 bytes (91.08% of a 3 MiB OTA slot), making firmware growth the tighter storage constraint.
- No board-level joules-per-inference or building-level electrical savings have yet been measured with calibrated power instrumentation.

PROJECT DEFINITION

2. Problem Statement, Objectives & Scope

2.1 Problem statement

Conventional classroom switching is typically manual or room-wide. This creates two inefficiencies: devices may stay powered after occupants leave, and devices serving unoccupied portions of a room may remain active because other zones are still in use. Visualec addresses the second problem by adding spatial awareness to the control loop.

2.2 Core objectives

1. Capture the classroom environment and obtain frames suitable for human occupancy detection.
2. Run compact INT8 person detection on an ESP32-S3-class edge target without relying on cloud inference.
3. Map detected people to predefined classroom zones and maintain zone-wise occupancy counts.
4. Apply deterministic energy-optimization rules and hysteresis to avoid rapid relay toggling.
5. Control zone-associated lights and fans through GPIO-driven relay channels with manual override and fail-safe behavior.
6. Expose occupancy, relay state, device health, logs, and energy observations through a local monitoring dashboard.
7. Validate inference performance, memory headroom, control response, and later physical energy savings using measured evidence.

2.3 Current prototype scope

In scope now	Outside current evidence / future scope
Three-zone smart-classroom control concept	Full building or utility-grid dispatch
ESP32-S3 INT8 person inference	Stock YOLOv8n inference directly on ESP32-S3
Zone-wise occupancy mapping	Final production accuracy of custom student model
3-channel relay actuation architecture	Certified mains switching hardware and installation
Local dashboard and runtime status	Cloud-scale fleet management
Software-derived energy observations	Power-meter-verified building energy savings

ARCHITECTURE

3. System Architecture

The architecture follows a closed-loop sequence: input -> processing -> AI/logic -> decision -> controller -> output -> monitoring. The attached project diagram consolidates the full flow from camera acquisition through ESP32-S3 processing, zone-wise control, relay actuation, and dashboard reporting.

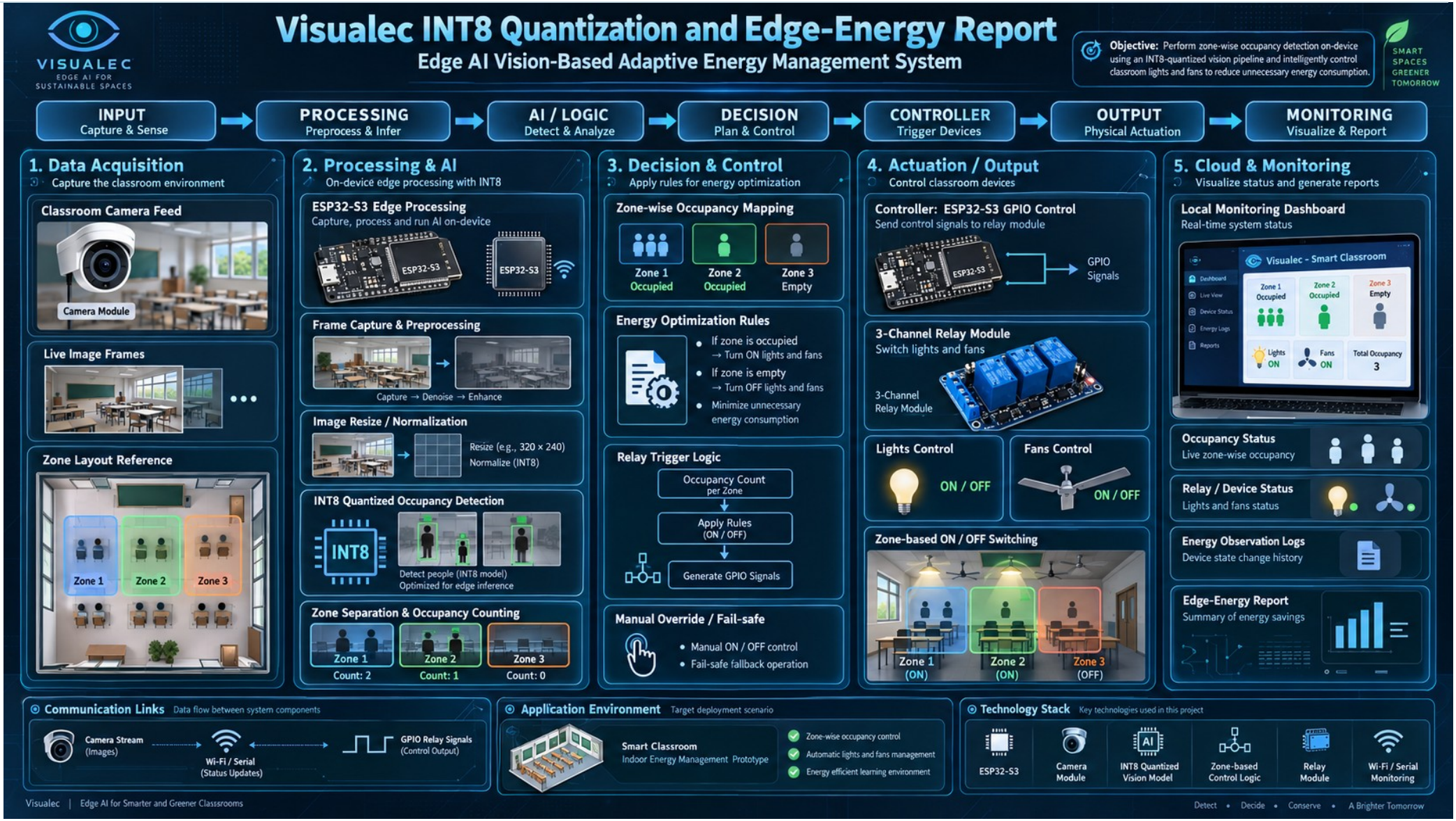


Figure 1. Visualec end-to-end architecture and workflow diagram provided with the project.

3.1 Architectural layers

Layer	Function	Primary elements
Data acquisition	Observe the classroom and retain zone layout context	Camera stream, live frames, predefined zone regions
Processing & AI	Prepare frames and infer people at the edge	ESP32-S3, resize/normalization, ESP-DL INT8 detector
Decision & control	Convert detections into stable zone decisions	Occupancy map, rule engine, timers/hysteresis, fail-safe
Actuation / output	Apply physical control actions	ESP32-S3 GPIO, 3-channel relay, lights, fans
Monitoring	Visualize and record system state	Local dashboard, occupancy/device status, logs, energy report

Architecture boundary

The diagram represents the intended integrated control architecture. The current technical evidence additionally states that the laptop still captures the C270 stream, performs motion selection and temporal tracking, and serves the dashboard; edge deployment therefore removes laptop neural inference, not all laptop functions.

DATA & CONTROL FLOW

4. Functional Workflow

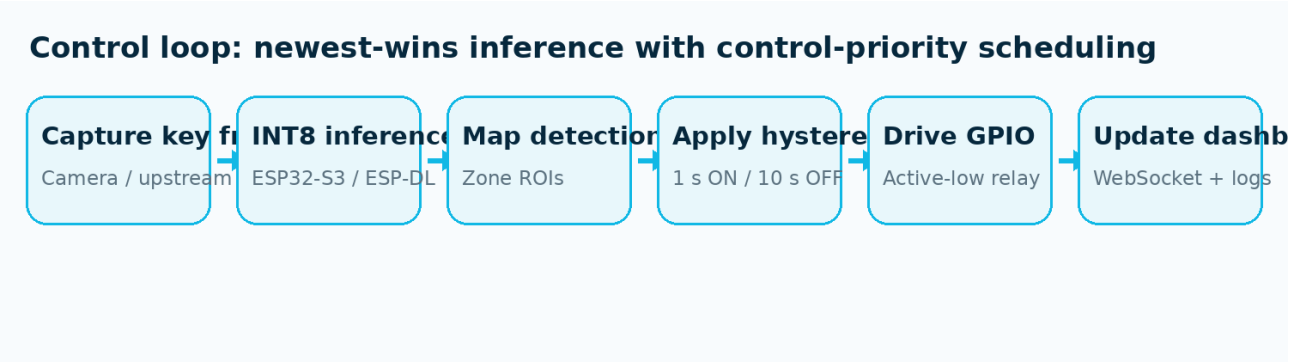


Figure 2. Runtime control loop and priority path.

4.1 Step-by-step execution

1. Capture or select the newest relevant camera frame and prepare it for transfer.
2. Resize and encode the wire frame as 288 x 160 RGB565 little-endian data for transport.
3. Run the compiled INT8 pedestrian detector on the ESP32-S3. The current bootstrap model has a 224 x 224 x 3 model input.
4. Filter detections using the configured confidence threshold, NMS, and top-K limits.
5. Assign each detection to the appropriate zone using the configured zone geometry and compute occupancy count per zone.
6. Apply activation/deactivation timing rules and convert stable zone occupancy into requested ON/OFF states.
7. Apply GPIO outputs to the relay channels, acknowledge the state only after the GPIO action has been applied, and publish status to the dashboard.
8. If camera/link state becomes stale, force a safe-off state according to the firmware fail-safe behavior.

4.2 Zone occupancy logic

For a predefined set of zone regions Z1...Zn, each valid person detection is associated with one zone using its spatial position in the frame. A zone is considered occupied when its validated count is greater than zero. The controller does not switch directly on every raw frame; it applies persistence timing so short detection gaps or transient false positives do not immediately toggle appliances.

Observed timing defaults

> The current firmware description reports default automatic activation and deactivation delays of approximately 1 s and 10 s respectively, with stale camera/link state forcing safe-off after approximately 30 s. These values should be tuned only after relay-response and user-comfort testing.

EDGE PLATFORM & ACTUATION

5. Hardware Design

5.1 Edge compute platform

Parameter	Configured / observed value
Device	ESP32-S3 N16R8
CPU target	Dual-core, 240 MHz
External memory	8 MiB OPI PSRAM at 80 MHz
Inference provider	ESP-DL INT8
Model partition	2 MiB dedicated partition
OTA application slot	3 MiB per slot
Relay behavior	Active-low: GPIO LOW = ON, GPIO HIGH = OFF

5.2 Sensing and controlled loads

- Camera / frame source: classroom vision stream, with the project pipeline referencing real Logitech C270 sessions for future training and calibration.
- Actuation: three relay channels in the prototype architecture, each associated with zone-specific electrical loads.
- Representative loads: lights and fans. The technical evidence uses three configured 9 W prototype loads for software energy arithmetic; these values are not power-meter measurements.
- User interface: local dashboard for occupancy, relay/device status, logs, and energy observations.

5.3 Hardware safety considerations

- The low-voltage ESP32-S3 control circuit must remain electrically isolated from any mains-voltage switching hardware.
- The prototype relay logic includes fail-safe OFF behavior when camera/link state is stale.
- Manual override is part of the control architecture and should remain available during testing and deployment.
- A production installation requires appropriately rated relay/contactor hardware, enclosure, fusing, grounding, and qualified electrical integration.

SOFTWARE STACK

6. Software & AI Architecture

Component	Role	Current evidence
ESP-IDF firmware	Task scheduling, GPIO, networking, configuration, metrics	Implemented in edge firmware
ESP-DL 3.3.11	INT8 graph execution on ESP32-S3	Deployed provider
pedestrian_detect 0.3.2	Bootstrap person detector component	Deployed model component
WebSocket / JSON	Persistent occupancy, commands, configuration and acknowledgements	Runtime communication path
RGB565 frame protocol	Transfers prepared frames to the edge device	288 x 160 RGB565LE with versioned header
Dashboard/backend	Monitoring, configuration, logs, energy observations	Part of current system architecture

6.1 Current detector configuration

Parameter	Value
Model	pedestrian-pico-bootstrap / PICO_S8_V1
Model input	224 x 224 x 3
Confidence threshold	0.45 (dashboard-configured)
NMS threshold	0.50
Top-K	10
Preprocessing	mean [0,0,0]; divisor [1,1,1]
Quantization	w8a8 signed INT8, symmetric power-of-two per tensor

Important model distinction

The current physical ESP32-S3 test does not run stock YOLOv8n. The running detector is the bootstrap ESP-DL model. YOLOv8n is planned as a teacher/pseudo-labeling model for training a smaller ESP-compatible student.

EDGE AI

7. INT8 Quantization & Model Pipeline

7.1 Why INT8 is used

INT8 stores each scalar in one byte rather than the four bytes used by FP32. When tensor count is held constant, the numeric representation therefore requires 25% of FP32 scalar storage. This helps fit model weights and intermediate activations within flash, PSRAM, caches, and working buffers, while also enabling integer-friendly execution on the ESP32-S3.

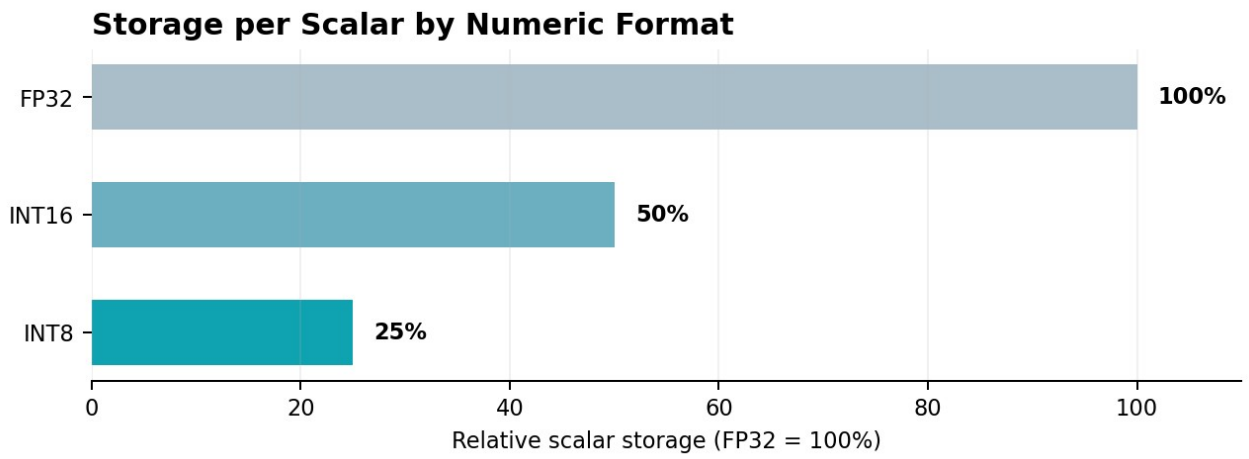


Figure 3. Relative scalar storage by numeric format. This is a representation comparison, not a latency or power guarantee.

7.2 Quantization mapping

The ESP-DL strategy described in the technical evidence uses symmetric, power-of-two, per-tensor quantization. For a floating-point value R, the signed 8-bit mapping is represented as: Q = clip(round(R / 2^Exp), -128, 127), with approximate reconstruction R_hat = Q x 2^Exp. Calibration selects exponents using representative real frames so typical activation ranges are covered.

Quantization trade-off

Values outside the selected range can saturate, while small numeric differences can collapse to the same integer after rounding. Representative calibration data is therefore critical to preserve person-detection quality.

7.3 Planned Visualec model-development sequence

1. Capture separate real Logitech C270 sessions for training, calibration, and held-out evaluation.

2. Use YOLOv8n as a person-only teacher to generate pseudo-labels on real frames.
3. Train a one-class ESPDet-Pico-derived student using the teacher detections.
4. Export the student model to ONNX.
5. Calibrate and quantize the student for ESP32-S3 with ESP-DL-supported operators.
6. Package and hash the resulting visualec_pico.espdl artifact.
7. Compare teacher, floating student, quantized host graph, and physical ESP32-S3 outputs.
8. Flash the custom model only after the defined accuracy and hardware gates pass.

> **PTQ before QAT**
> The intended pipeline uses post-training quantization first. Quantization-aware training is reserved for cases where the quantized student loses more than three percentage points against the floating student. Synthetic frames are not part of the stated training pipeline.

REAL-TIME CONTROL

8. Firmware Scheduling, Safety & Control Logic

8.1 Task separation

The edge firmware separates relay control, inference, metrics, networking, and configuration work. Relay control is assigned higher priority than inference so appliance state transitions are not blocked by a long inference call. The image pipeline uses a one-frame newest-wins queue, discarding stale keyframes rather than allowing a backlog to accumulate.

Task / path	Design intent
Relay control	Highest practical control priority; deterministic GPIO application
Inference	Runs on newest available keyframe; stale frames may be dropped
Networking	Maintains persistent edge-backend exchange
Metrics	Publishes device performance and health measurements
Configuration	Applies dashboard-controlled thresholds and runtime settings
Fail-safe	Forces safe-off when camera/link freshness conditions are violated

8.2 Relay trigger logic

The logical control state is derived from stable zone occupancy rather than a single frame. An occupied zone requests the associated light/fan channel to be ON, while an empty zone requests OFF after the configured deactivation delay. The controller generates GPIO signals only after applying timing and safety rules. Relay acknowledgements are issued after the physical GPIO state is applied, allowing the dashboard to distinguish commands from confirmed device state.

8.3 Manual override and fail-safe

- Manual ON/OFF control is retained for operator intervention and testing.
- Active-low output semantics must be handled consistently across initialization, normal operation, restart, and fail-safe paths.
- Safe-off behavior is used for stale camera/link conditions to avoid leaving loads energized indefinitely when visibility is lost.

CONNECTIVITY

9. Communication & Monitoring

9.1 Frame transport

Property	Value
Wire frame	288 x 160 RGB565 little-endian

Property	Value
Pixel payload	92,160 bytes
Versioned header	28 bytes
Total packet	92,188 bytes
Purpose	Transport prepared image data to the edge inference path

9.2 Control and telemetry

Occupancy results, commands, configuration, and status data are exchanged as JSON over a persistent WebSocket connection. This channel supports dashboard-controlled thresholds, live occupancy state, relay/device acknowledgement, and runtime metrics.

9.3 Dashboard responsibilities

- Show live zone-wise occupancy and total detected occupants.
- Show requested and confirmed relay/device state for lights and fans.
- Store device-state changes and operational events in logs.
- Expose configurable detection thresholds and operating parameters.
- Summarize software-derived energy observations while clearly labeling them as estimates until physical metering is added.

MEASURED EVIDENCE

10. Performance Validation

The current evidence is a physical ESP32-S3 verification sample. It demonstrates that real frames reach the edge device, inference results are produced, frame IDs continue to advance, and the runtime protocol operates without errors during the observed interval.

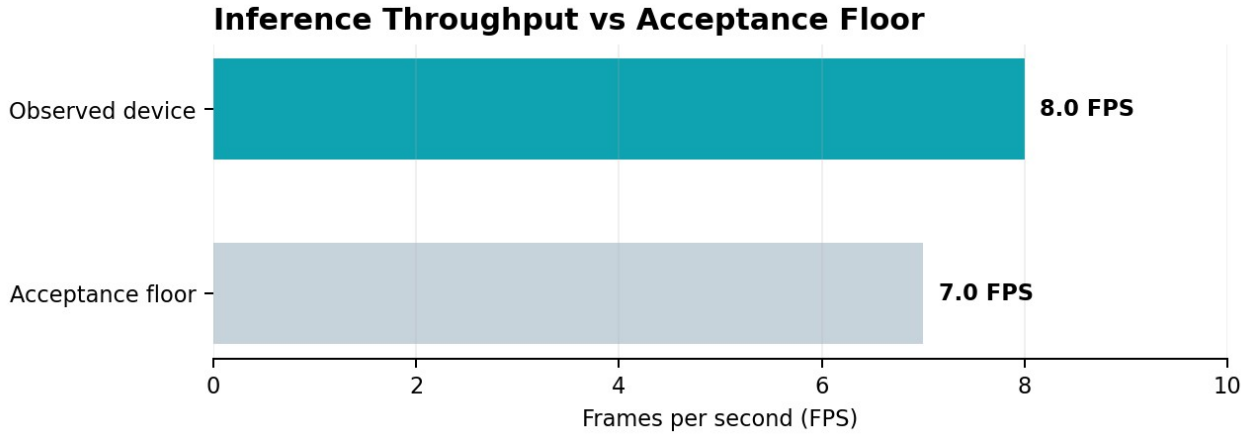


Figure 4. Observed device throughput passes the 7 FPS acceptance floor.

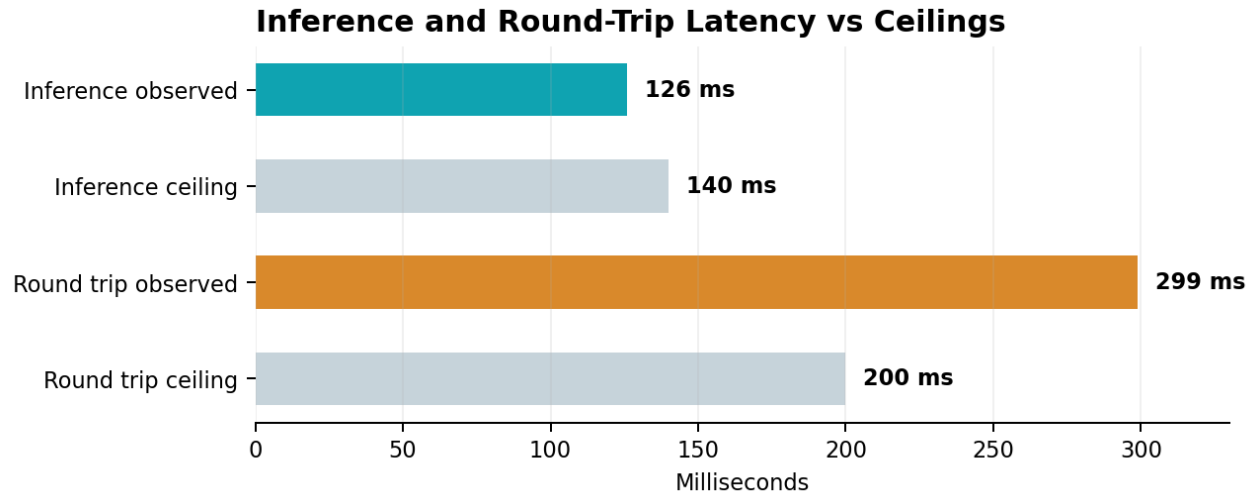


Figure 5. Device inference passes its ceiling; round-trip latency remains above the p95 target.

Metric	Observed	Acceptance / reference	Status
Device inference	126 ms	<= 140 ms	PASS
Compute capacity	8 FPS	>= 7 FPS	PASS
Round-trip latency	299 ms	<= 200 ms p95 target	NOT MET
Protocol errors	0	0 preferred	PASS in observed sample
Current people count	1	Context only	Observed
Frame IDs	237 -> 247	Continuous advancement	Observed
Completed frames	234 -> 244	Continuous result flow	Observed

Primary performance gap

The edge compute gate is currently satisfied. The 299 ms observed round trip is about 99 ms above the 200 ms target, so transfer, queueing, and backend handling should be profiled before further increasing model complexity.

RESOURCE UTILIZATION

11. Memory & Partition Analysis

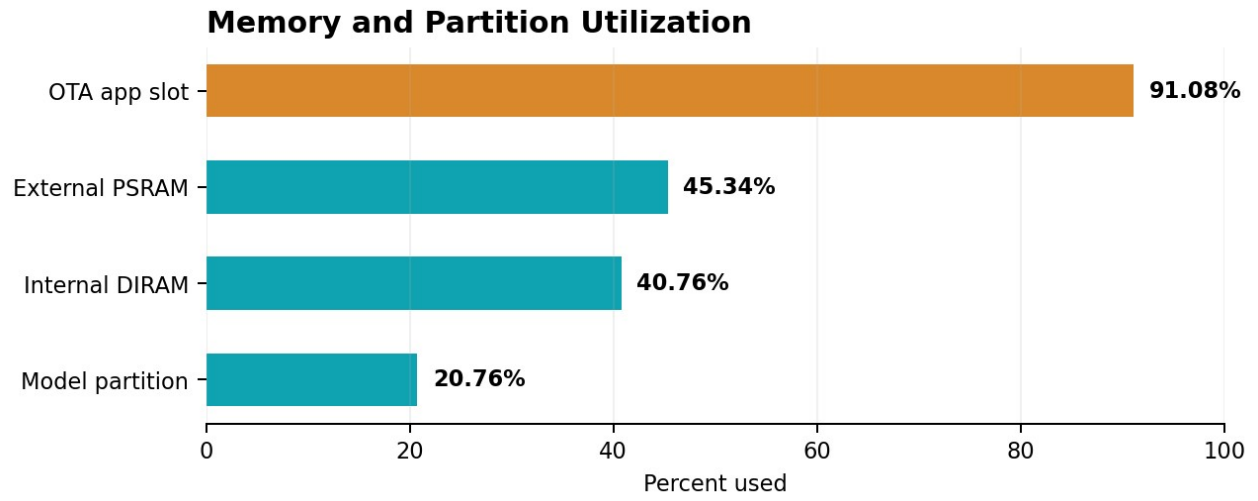


Figure 6. Built-artifact and retained runtime memory utilization.

Resource	Capacity	Used	Free	Used %
OTA application slot	3.15 MB	2.87 MB	~280.46 KB	91.08%
External PSRAM	8.39 MB	3.80 MB	~4.58 MB	45.34%

Resource	Capacity	Used	Free	Used %
Internal DIRAM	341.76 KB	139.31 KB	202.45 KB	40.76%
Model partition	2.10 MB	435.44 KB	~1.66 MB	20.76%

The dedicated model partition has substantial headroom, but the application OTA slot is tight. With the current binary already using 91.08% of a 3 MiB slot, additional firmware features, libraries, debugging code, or model-wrapper expansion can become a deployment constraint before model storage itself does.

Growth priority

Optimize firmware and dependency footprint before assuming that the remaining flash is dominated by the model. The custom model may fit comfortably in the model partition while the application image still exceeds the OTA budget.

ENERGY EVIDENCE

12. Energy Management & Measurement Methodology

12.1 Control-side energy model

Visualec saves load energy by keeping a controlled device OFF during verified periods when its zone does not require service. For controlled load i, the avoided energy is calculated as E_saved,i = P_i x verified_off_time_i. Across all zones, the load-side savings are the sum of these avoided-energy terms.

The current prototype configuration uses three 9 W loads in its software arithmetic. If all three are OFF, the instantaneous avoided load power in the model is 27 W. This is configuration arithmetic only; it is not a power-meter observation and should not be presented as a measured saving.

12.2 Edge-compute energy model

INT8 can reduce data movement and integer arithmetic cost relative to larger floating-point representations, but whole-board energy depends on more than model arithmetic. Board regulators, PSRAM, Wi-Fi duty cycle, USB circuitry, relay coils, frame transfer, and firmware behavior all contribute. The current evidence therefore does not convert INT8 storage reduction directly into joules-per-inference.

12.3 Required measurement procedure

1. Insert a USB power meter or calibrated shunt between the supply and ESP32-S3 board and record voltage/current for idle, connected/no-inference, and sustained high-motion inference states.
2. Record inference start/end markers and calculate joules per inference as integral(V x I dt) divided by completed inferences over at least 1,000 real frames.
3. Measure laptop wall power with local YOLO inference enabled and disabled while capture, tracking, and dashboard activity remain identical.
4. Meter each controlled load or use verified nameplate power, then compute avoided energy from physically verified relay-state durations.
5. Capture separate real C270 train, calibration, and held-out sessions; train and quantize the custom student and enforce the existing quality gates.
6. Report total-system energy as camera + laptop + ESP board + relay modules + controlled loads rather than reporting ESP consumption alone.

Claim boundary

The current system proves edge execution and control feasibility. It does not yet prove electrical energy savings, joules-per-inference, or final custom-model recall.

ENGINEERING READINESS

13. Implementation Status, Risks & Limitations

13.1 Current status

Area	Status	Evidence / next step
Physical ESP32-S3 INT8 inference	Implemented	126 ms observed inference; 8 FPS compute capacity
Bootstrap pedestrian detector	Implemented	pedestrian-pico-bootstrap / PICO_S8_V1
Zone-based control architecture	Implemented / integrated concept	Decision, GPIO, relay and dashboard flow defined
Custom Visualec student model	Planned	Train real-data ESPDet-Pico-derived student
Round-trip latency target	Needs optimization	299 ms observed vs 200 ms target
Energy savings measurement	Not yet measured	Add calibrated board and load power metering
Final accuracy / recall evidence	Not yet available	Held-out real dataset and hardware agreement tests required

13.2 Main risks and mitigations

Risk	Impact	Mitigation
Round-trip latency remains high	Delayed dashboard/control feedback	Profile packet transfer, queueing and backend processing; preserve newest-wins behavior
OTA slot nearly full	Firmware growth can fail to build/flash	Remove unused components, optimize logging, review partition strategy
Calibration set is not representative	INT8 recall degradation	Use real C270 frames spanning lighting, occlusion and seating conditions
Relay chatter from unstable detections	Device wear and poor user experience	Use activation/deactivation persistence and minimum hold times
Network/camera loss	Loads may remain ON incorrectly	Maintain stale-state safe-off and manual override
Software energy estimates mistaken for measurements	Overstated project impact	Label estimates clearly; add power instrumentation before external claims

NEXT PHASES

14. Scalability & Future Roadmap

14.1 Near-term engineering roadmap

1. Reduce round-trip latency below the 200 ms target by profiling frame transfer, queueing, and backend paths.
2. Train visualec_pico.espdl from real C270 data using the teacher-student sequence and validate hardware agreement.
3. Add calibrated current/voltage measurement for the ESP32-S3 and controlled loads.
4. Run long-duration classroom tests to quantify false switching, occupancy recall, relay duty cycles, and comfort impact.
5. Optimize OTA application footprint and reserve explicit margin for future firmware features.
6. Add formal fault-injection tests for stale links, camera loss, malformed packets, restart behavior, and manual override.

14.2 Scaling beyond the three-zone MVP

- Increase the number of logical zones while preserving edge inference and low-latency local control.
- Use distributed ESP nodes or a hierarchical room controller for larger classrooms and multi-room deployments.
- Integrate direct energy metering to convert relay-state duration into measured energy baselines and verified savings.
- Aggregate room-level telemetry into a building energy-management dashboard while keeping safety-critical switching local.

- Add occupancy trends, scheduling context, and maintenance analytics only after the core person-detection and control loop remains reliable.

14.3 Production-readiness gates

Gate	Required evidence
Accuracy	Held-out real-room person recall/precision and zone-assignment correctness
Latency	Inference and end-to-end p95 targets under sustained traffic
Energy	Measured board energy per inference and measured load energy before/after control
Reliability	Long-duration stability, fail-safe recovery, relay-cycle testing
Electrical safety	Properly rated isolated switching hardware and qualified installation
Maintainability	OTA margin, configuration management, logs, reproducible model artifacts and hashes

IMPACT FRAMING

15. Sustainability Alignment

Visualec is intended to reduce avoidable electrical consumption by matching device operation more closely to actual occupied zones. At the current prototype stage, this sustainability value is architectural rather than quantified: the project demonstrates the mechanism for adaptive switching, but verified energy savings require the metering procedure defined in Section 12.

Sustainability theme	Project contribution
Energy efficiency	Reduces unnecessary ON-time of zone-controlled lights and fans
Smart infrastructure	Uses edge vision and local automation for responsive room control
Responsible resource use	Supports evidence-based energy observation and device-state logging
Low-latency local intelligence	Keeps core occupancy-to-relay decisions close to the device instead of depending on cloud inference

SDG relevance (conceptual)

The project concept is broadly aligned with SDG 7 (Affordable and Clean Energy), SDG 9 (Industry, Innovation and Infrastructure), SDG 11 (Sustainable Cities and Communities), SDG 12 (Responsible Consumption and Production), and SDG 13 (Climate Action). Quantified impact should be tied to measured energy data rather than inferred from the SDG labels alone.

SUMMARY

16. Conclusion

Visualec demonstrates a practical architecture for zone-aware classroom energy control using compact edge vision. The current ESP32-S3 evidence confirms that INT8 pedestrian inference is running on physical hardware and that the device compute target is met at approximately 8 FPS and 126 ms inference latency. The system architecture then converts occupancy into stable GPIO-controlled relay actions and publishes status through a local monitoring layer.

The strongest remaining engineering issues are not model storage but end-to-end latency, OTA application headroom, final custom-model validation, and physical energy measurement. The project should therefore progress by optimizing the 299 ms round trip, training the real-data compact student, instrumenting actual board and load power, and validating the complete control loop over long classroom sessions. With those gates satisfied, Visualec can move from a credible edge-AI prototype to a defensible energy-management system with measured operational benefit.

APPENDIX

Appendix A - Verification Matrix

ID	Verification item	Target / criterion	Current result
V-01	Physical INT8 inference executes	Successful inference on ESP32-S3	PASS
V-02	Device inference latency	<= 140 ms	126 ms - PASS
V-03	Compute throughput	>= 7 FPS	8 FPS - PASS
V-04	End-to-end round trip	<= 200 ms p95 target	299 ms - NOT MET
V-05	Protocol integrity	No protocol errors in sample	0 errors - PASS
V-06	Continuous result flow	Frame IDs and completed frames advance	Observed 10-frame advancement
V-07	Custom Visualec model	Trained and deployed artifact	NOT YET
V-08	Hardware accuracy/recall	Held-out real dataset gates met	NOT YET
V-09	Board energy per inference	Calibrated joules/inference	NOT YET
V-10	Verified load savings	Metered before/after energy	NOT YET

Recommended next test campaign

- 1,000+ real-frame energy test with synchronized inference markers.
- Low-light, partial-occlusion, group seating, empty-room, and entry/exit sequences.
- Long-duration run to quantify false ON/OFF transitions per hour.
- Network degradation and disconnect tests to confirm stale-state safe-off.
- Cold boot, warm restart, and relay initialization tests for active-low outputs.
- Teacher vs floating student vs quantized host vs physical ESP detection agreement report.

APPENDIX

Appendix B - Key Parameters & Glossary

B.1 Key parameters

Parameter	Value
Device	ESP32-S3 N16R8, dual-core 240 MHz
PSRAM	8 MiB OPI PSRAM at 80 MHz
ESP-DL runtime	3.3.11
Pedestrian component	pedestrian_detect 0.3.2
Deployed model	pedestrian-pico-bootstrap / PICO_S8_V1
Model input	224 x 224 x 3
Wire frame	288 x 160 RGB565LE
Packet size	92,188 bytes (28-byte header + 92,160-byte pixels)
Confidence / NMS / top-K	0.45 / 0.50 / 10
Quantization	w8a8 signed INT8, symmetric power-of-two per tensor
Model artifact size	435,440 bytes
Application binary	2,865,264 bytes

B.2 Glossary

Term	Meaning
ESP-DL	Espressif deep-learning inference framework for ESP devices
INT8	Signed 8-bit integer representation used for quantized model

Term	Meaning
	weights/activations
PTQ	Post-training quantization
QAT	Quantization-aware training
NMS	Non-maximum suppression for filtering overlapping detections
FPS	Frames processed per second
Round trip	End-to-end time covering frame/result transport and backend path
PSRAM	External pseudo-static RAM used for larger buffers
DIRAM	Internal data/instruction-capable RAM region
OTA slot	Flash partition used for over-the-air application images
Zone ROI	Predefined region of interest used to map person detections to a classroom zone

SOURCE BASIS

References

1. Visualec INT8 Quantization and Edge-Energy Report, technical evidence snapshot dated 06 September 2026 (provided project source).
2. Visualec system architecture / workflow infographic supplied with this report.
3. Visualec live system-status snapshot and system-status API tables referenced by the supplied technical evidence.
4. Visualec edge firmware and model pipeline: ESP-IDF configuration, ESP32-S3 runtime, ESP-DL model wrapper, and real-frame model pipeline referenced by the supplied technical evidence.
5. Espressif ESP-DL quantization documentation, referenced in the supplied technical evidence for the w8a8 symmetric power-of-two quantization scheme.

Report integrity note

Performance, memory, model, and energy-claim boundaries in this report are derived from the supplied Visualec evidence. Proposed scale-up items are explicitly presented as future work rather than completed results.